



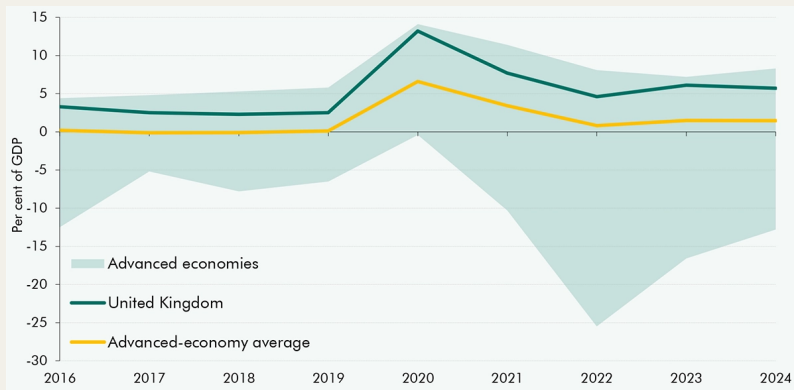
How Much Government Debt is Too Much?

UNIQ - Economics and Management

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Advanced economies are, on average, borrowing more now than in recent history



GREEK SOVEREIGN DEBT CRISIS, 2010–2012

What we *don't* want to happen

The fiscal deficit of Greece worsened significantly during the 2008–09 financial crisis, when the economy also dipped into a recession. The cost of borrowing for Greece started to rise, and they could not refinance their existing debts.

2010 A **€110bn** emergency bailout to avert default and stop contagion to Spain and Portugal.

2011 A **50% reduction** (haircut) on privately-held bonds, aiming to drag debt down to $\approx 120\%$ of GDP.

Debt sat near 100% of GDP for two decades — then, within months, Greece was locked out of financial markets entirely.

A TALE OF TWO DEBT BURDENS

Why is Argentina forced to slash public spending, whereas Japan plans on increasing it?



Public debt-to-GDP

(2023) 158% → 82%* (predicted 2026)



Public debt-to-GDP

(2020) 258% → 249% (2025)

GROWTH PATHS

The Debt Sustainability Equation

$$\underbrace{\Delta b_t}_{\text{change in the debt-to-GDP ratio}} = d_t + (i_t - \pi_t - g_t) \underbrace{b_{t-1}}_{\text{debt stock}}$$

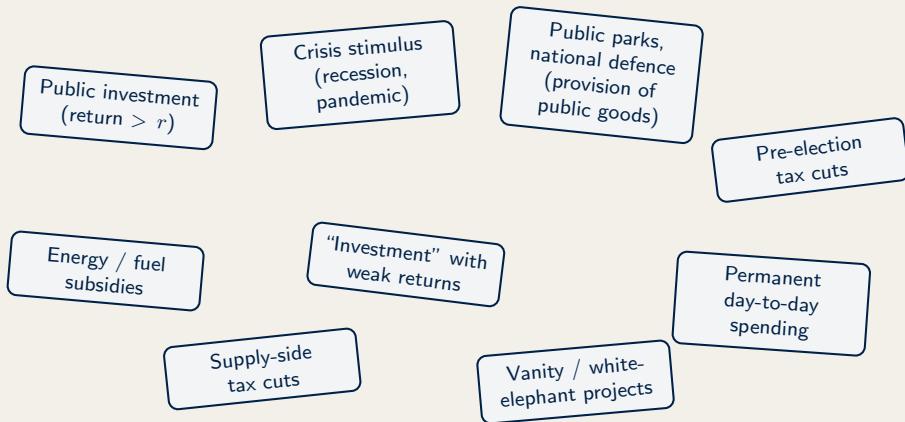
$$\Delta b_t = \underbrace{d_t}_{\text{primary balance}} + (i_t - \pi_t - g_t) b_{t-1}$$

The *primary balance* d_t tracks how much a government spends versus what it raises, *before* interest costs. A primary deficit ($d_t > 0$), where spending exceeds gov. revenue, adds directly to the debt.

Why don't governments just always run a primary surplus ($d_t < 0$)?

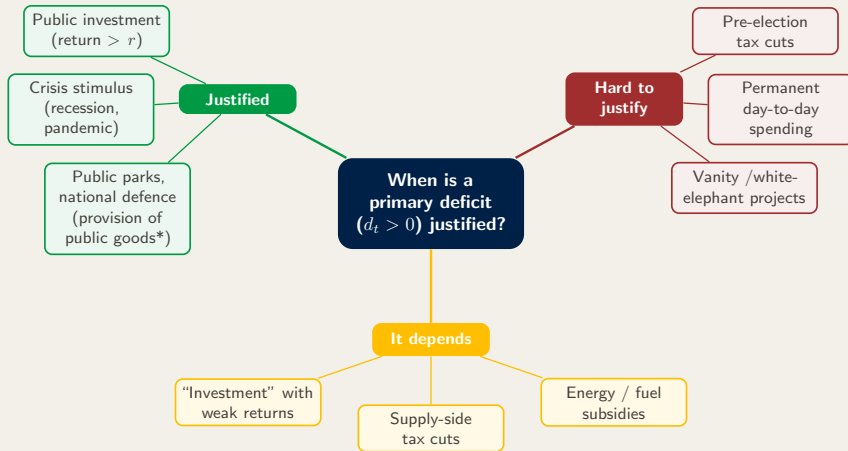
THE PRIMARY BALANCE

Why do sovereigns borrow?



THE PRIMARY BALANCE

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THE PRIMARY BALANCE

Valid Public Expenditure

Efficiency: does it raise future growth g , or smooth a *temporary* shock?

Public goods: does it provide a good that can be:

1. provided to everyone without additional cost, and;
2. cannot (feasibly) be excluded from anyone?

Permanent consumption: if a country is routinely spending more on permanent, predictable consumption (pensions, health care), the gap must be closed to prevent the debt stock rising year-on-year

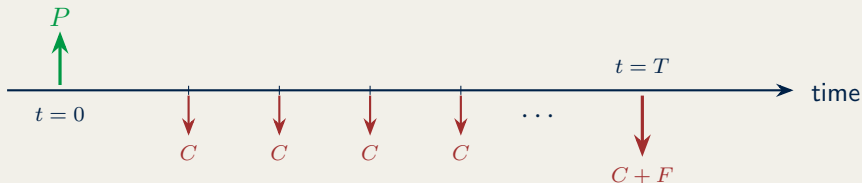
$$\Delta b_t = d_t + \left(\underbrace{i_t}_{\text{interest rate}} - \pi_t - g_t \right) b_{t-1}$$

The nominal *interest rate* i_t sets the price of servicing debt. What determines the rate a government is charged to borrow?

INTEREST RATES

What determines the price of borrowing for a Government?

Governments fund their expenditure via financial markets by issuing **bonds**, a contract that promises a payment (the face value F) at some date in the future (the maturity date), with regular fixed payments in between (coupons C) for some upfront price P



The price of this contract depends on how much investors value these regular payments, adjusted for any risk. **Question:** is this contract risky, or risk-free?

INTEREST RATES

What determines the price of borrowing for a Government?

A bond's **yield** i is the return an investor earns from a bond. It is expressed as a percentage of its current market price, so price P and yield i are inversely related. If more perceived risk $\Rightarrow$ investors pay a lower price $P \Rightarrow$ a higher yield i .

$$i \approx \text{expected policy rate} + \text{term premium} + \text{risk premium}$$

- **Expected policy rate:** the short-term rates which investors expect the central bank to set over the bond's life (this reflects expected inflation and growth).
- **Term premium:** extra compensation for investors holding bonds *long*: money is tied up while rates and inflation may move.
- **Risk premium:** extra return for the chance of *not being repaid* (default).

Further reading: CORE Econ: rates of return, risk & bond prices

$$\Delta b_t = d_t + \underbrace{(i_t - \pi_t - g_t)}_{r_t - g_t \text{ (snowball)}} b_{t-1}$$

Write the real rate $r_t = i_t - \pi_t$, and the bracket becomes $r_t - g_t$. When the real interest rate *exceeds* the growth rate g_t , debt grows under its own weight – even with a balanced primary budget.

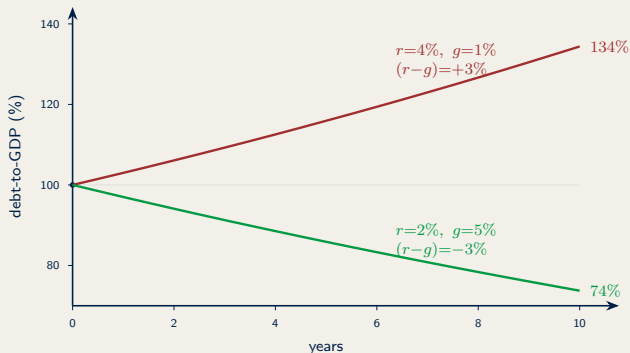
$$r > g$$

Why is debt-to-GDP important?

$$\Delta b_t = \underbrace{d_t}_{\text{primary balance the level}} + \underbrace{(r_t - g_t) b_{t-1}}_{\text{snowball the growth}}$$

- **Level (d_t):** a proportion of GDP added every year, **independent of the stock.**
- **Growth ($(r_t - g_t)b_{t-1}$):** scales with the stock, so it *compounds*.

Set $d_t = 0 \rightarrow$ the path is a race between r and g .



$$r > g$$

Why is debt-to-GDP important?

Economy	Debt %GDP	r %	g %	$r - g$ pp = % of debt	stabilising primary balance % of GDP
India	83	1.5	6.5	-5.0	-4.2
China	90	2.0	4.8	-2.8	-2.5
Japan	235	-1.0	1.0	-2.0	-4.7
USA	120	1.0	2.0	-1.0	-1.2
Germany	63	0.5	0.2	+0.3	+0.2
UK	100	2.0	1.3	+0.7	+0.7
Italy	137	2.5	0.5	+2.0	+2.7
Argentina	80	9.0	4.0	+5.0	+4.0

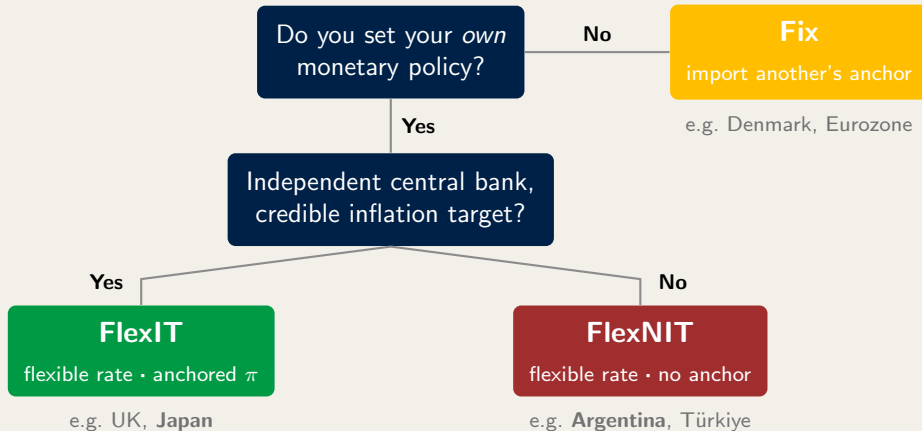
Illustrative ≈ 2025 : g = real GDP growth (IMF WEO); r = real effective rate on government debt (rounded).

$$\Delta b_t = d_t + (i_t - \underbrace{\pi_t}_{\text{inflation}} - g_t) b_{t-1}$$

Inflation π_t erodes the real value of outstanding debt. The monetary regime that sets π_t , and how it interacts with fiscal policy, defines the menu of policy responses available that propagate through every other term.

MONETARY-FISCAL INTERACTIONS

Two questions decide how a country's inflation gets anchored



MONETARY-FISCAL INTERACTIONS

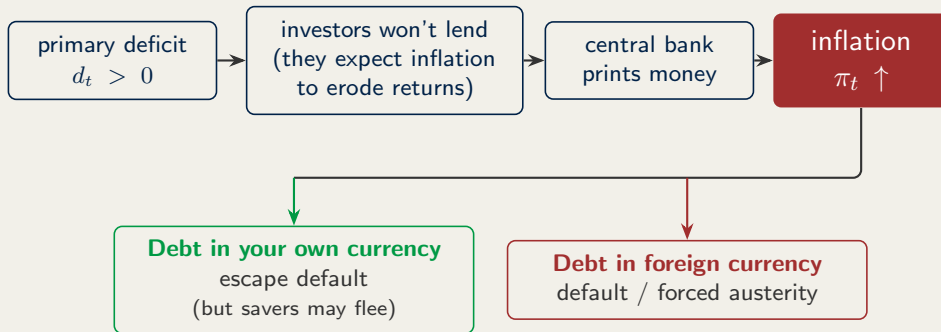
What each regime does

	FlexIT	FlexNIT	Fix
Who anchors inflation?	the central bank	nothing firm	the anchor country
The exchange rate	floats	floats, often sliding	held fixed
After a shock, the exchange rate...	adjusts to cushion debt growth	adjusts to amplify debt growth	cannot adjust

The exchange rate is a shock-*absorber* in FlexIT, a shock-*amplifier* in FlexNIT.

MONETARY-FISCAL INTERACTIONS

Deficits under a non-credible central bank



MONETARY-FISCAL INTERACTIONS

How the regime reaches every term

$$\Delta b_t = d_t + (i_t - \pi_t - g_t) b_{t-1}$$

π_t FlexIT anchors it low • FlexNIT lets it drift • Fix imports it.

i_t credibility shrinks the risk & term premium $\Rightarrow$ lower i_t .

g_t unstable inflation and uncertain pegs drag on growth.

Real rate $r_t = i_t - \pi_t$: a burst of inflation shrinks debt *once* $\rightarrow$ but markets then demand a higher i_t . **No free lunch.**

MONETARY-FISCAL INTERACTIONS

Japan vs Argentina

Japan — FlexIT

own currency (the yen), credible
Bank of Japan
inflation anchored $\rightarrow i_t$ tiny
carries $\approx 250\%$ debt

Argentina — FlexNIT

anchor never held $\rightarrow$ high, volatile
inflation
savers fled to dollars $\rightarrow$ dollarised
can't print dollars $\rightarrow$ forced austerity

$$\Delta b_t = d_t + (i_t - \pi_t - g_t) \underbrace{b_{t-1}}_{\text{existing debt}}$$

Last period's *stock* of debt, b_{t-1} . Its magnitude matters, but so does *who holds it* and *in which currency* it is owed.

Borrow at home or abroad?

Issuing in Own currency (e.g. ¥, £, \$)

Why it stabilises

- **Exchange-rate mitigation:** no currency mismatch → a fall in the exchange rate does *not* raise the real debt burden.
- **No forced default:** in extremes the central bank can always create the currency to pay (cost shows up as inflation, not a missed payment).

The catch: liquidity constraints

- If deep domestic demand from savers (banks, pension funds, insurers) → demand is sticky and slow to flee to other assets → lower yields.
- Low domestic demand from savers → higher yields and crowding out of private borrowers.

Borrow at home or abroad?

Issuing in Foreign currency (usually \$)

Why it tempts

- **Greater liquidity:** access to deep global capital markets and a far larger investor pool than a shallow home market can offer.
- Often a *lower headline rate*.

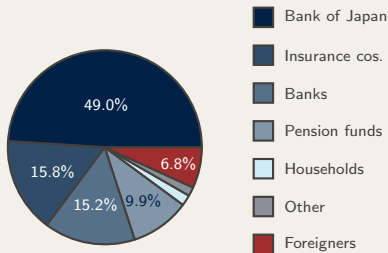
The catch: “the original sin”

- **Exchange-rate risk:** a depreciation of a country's domestic currency against the dollar will inflate the real value of the debt.
- **Balance-of-payments risk:** servicing requires *earning or buying foreign exchange* (exports, reserves, a lender).
- **Rollover / sudden-stop risk:** foreign creditors can flee at once → most likely during a crisis, precisely when it is needed most.

Who owns it? Japan vs Argentina

Japan — JGBs

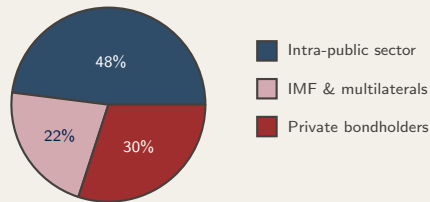
% of JGBs outstanding



≈100% in yen. Half is owed to the central bank;
foreigners hold ≈7%.

Argentina — gross public debt

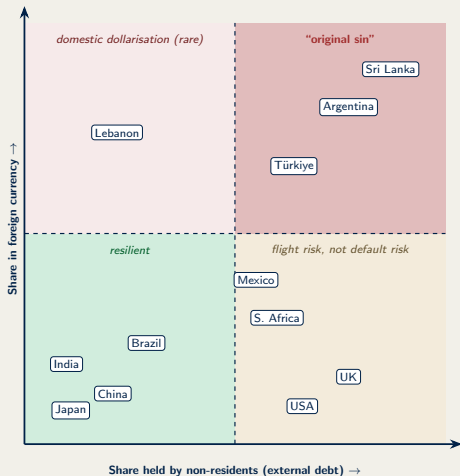
by creditor



≈2/3 in foreign currency. It also borrows heavily “from
itself” — but in dollars it cannot print.

Same trick (borrowing from the public sector), opposite outcome — because of the *currency*, not the ratio.

Who lends, and in whose currency



BRINGING EVERYTHING TOGETHER

The debt sustainability equation in full

$$\Delta b_t = d_t + (i_t - \pi_t - g_t) b_{t-1}$$

Δb_t change in the debt-to-GDP ratio in period t

b_{t-1} debt-to-GDP ratio at the end of the previous period

d_t primary deficit (spending – revenue, before interest)

i_t nominal interest rate on government debt

π_t inflation rate

g_t real GDP growth rate

The Chancellor's Game

